

# Re-analysis Summary Report for Peer-evaluation

*Paper ID: Anderson\_AmEcoJourn\_2011\_bLe8*

*Re-analyst's ID: fr167*

## Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

*For both tasks, you should provide your evaluation via this survey:* <https://forms.gle/EAREaC6JjN2PWFfR6>

## The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:

... [the study] finds substantially higher income for low-caste households residing in villages dominated by BACs [compared to villages dominated by upper caste]. (p. 240.)

## The corresponding article

Here you can find the original article: [https://osf.io/hcxdq/files/osfstorage/5e56aac535d22d02f75f482c?view\\_only=32ccf80d2c684ea9ac10ebbc4ec34df6](https://osf.io/hcxdq/files/osfstorage/5e56aac535d22d02f75f482c?view_only=32ccf80d2c684ea9ac10ebbc4ec34df6)

and the original data: [https://osf.io/k7xzy/?view\\_only=c1f2b9aaf05d486c942783efd8ec57ec](https://osf.io/k7xzy/?view_only=c1f2b9aaf05d486c942783efd8ec57ec)

They are provided only as context of the re-analysis.

**In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:**

Statistical analyses In the first step, the main file ("household.dta") and summary data ("table2.dta") were imported into R from the target paper (Anderson, 2011). It became evident that the two datasets differ in the number of observations. There are two fewer households in the main file compared to the summary from the table2 file. However, checking the results with the additional two households did not change the patterns of results. Thus, the data from the main file was used in further analyses. Next, the assumptions of normality were tested regarding the variable of interest (the total income). The normality was assessed following the recommended criteria. The skewness values above |2| and kurtosis values above |7| were treated as potentially violating the assumptions of normality (Kim, 2013). Finally, the Mahalanobis Distance was calculated relying on the usually recommended cutoff (i.e., < 0.001; (Mahalanobis, 1960; Penny, 1996) when screening for potential outliers. Multilevel analysis with a maximum likelihood estimator was then conducted. Households were nested within villages to account for the non-independence between the household of the same villages. Then, the outcome variable was grand-mean centred. In the multilevel model, standardised total income was regressed on a dichotomous predictor variable, depicting whether the given village was

dominated by low (coded as 1) or high (coded as 0) caste. As a robustness check, the more traditional test was also used to test for the above relationship (i.e., a linear regression analysis). The pattern results remained the same. Hence, herein are presented the results of a more advanced approach, that is, the multilevel analysis. All the analyses were performed in R (4.1.2). The list of used packages included: haven (Wickham, Miller, et al., 2022), dplyr (Wickham, François, et al., 2022), parameters (Lüdtke et al., 2020), lme4 (Bates et al., 2015), merTools (Knowles & Frederick, 2020), ggplot2 (Wickham, 2016), extrafont (Chang, 2022), cowplot (Wilke, 2020), mitml (Grund et al., 2021), lattice (Sarkar, 2008), e1071 (Meyer et al., 2021).

**Results** The analysis of skewness provided evidence that the assumptions of normality were violated regarding the outcome variable (total income), indicating that there might be potential outliers. Based on the Mahalanobis distance, data from 22 households were flagged as potential outliers. However, the analysis with and without these data did not change the results. Hence, the full dataset was retained. After investigating the villages' residuals (the joint distribution of the random intercepts and slopes), it became evident that six villages dominated by low-caste (i.e., villages: 70, 61, 60, 84, 59, and 58) violated the assumption of normality in the multivariate group-level distribution of the residuals (Jones & Subramanian, 2019). Hence, the analyses were repeated with and without the outlying villages. The results of the multilevel models are presented in Table 1. Households from low-caste dominated villages had higher total income than households from high-caste dominated villages. However, this effect disappeared when using data without the six outlying low-caste dominated villages (see Table 1). Thus, it seems that the six (6.67%) villages (out of 90 analysed) with 84 (6.49%) households (out of 1295 households) carry the positive relationship between the households' income and village's low-caste domination. The mean of the total households' income in low-caste dominated villages in the dataset without the six villages is 4925.46 (SD = 9676.15), while across the six villages 26787.4 (SD = 33073.85). Figure 1 presents the striking differences with and without the outlying villages. Great caution is warranted in drawing definite conclusions regarding the link between the households' income and caste domination on the village level.

**Table 1** Results of the multilevel linear models regressing households' total income on the type of village (i.e., whether it is low or high-caste dominated, with low-caste coded as one and high-caste coded as zero) with households nested within villages (left-pane presents results of the model performed on the full dataset, and right-pane presents results of the model performed on the dataset without six outlying countries). Full dataset Dataset without six outlying villages

|                                      | Fixed effect | SE    | 95% CI         | p        | SE    | 95% CI                | p     |
|--------------------------------------|--------------|-------|----------------|----------|-------|-----------------------|-------|
| High vs low village caste domination | 0.172        | 0.052 | [0.070, 0.273] | < 0.001* | 0.068 | 0.035 [-0.001, 0.137] | 0.052 |
| Random effect Variance               |              |       |                |          |       |                       |       |
| SD                                   |              |       |                |          |       |                       |       |

Intercept 28847435 5371 2110751 1453

Note. \* $p < 0.001$ . BIC for the full dataset = 28056.3, AIC for the full dataset = 28035.7, BIC for the dataset without outlying villages = 25316.5, AIC for the dataset without outlying villages = 25296.1, variance of the residuals for the full dataset = 133781622, SD = 11566, variance of the residuals for the dataset without outlying villages = 66914862, SD = 8180, ICC for the full dataset = 0.177, ICC for the dataset without outlying villages = 0.031,  $r^2$  for the full dataset = 0.029,  $r^2$  for the dataset without outlying villages = 0.005, dfresiduals for the full dataset = 1291, dfresiduals for the dataset without outlying villages = 1207, deviance for the full dataset = 28027.7, deviance for the dataset without outlying villages = 25288.1.

**Figure 1** A comparison of means and standard deviations of households' total income across villages dominated by high versus low-caste on the full dataset (left-pane) and dataset without six outlying villages (right-pane).

Note. Error bars represent 95% confidence intervals.

**References** Anderson, S. (2011). Caste as an Impediment to Trade. *American Economic Journal: Applied Economics*, 3(1), 239–263. <https://doi.org/10.1257/app.3.1.239> Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01> Chang, W. (2022). extrafont: Tools for Using Fonts. R package version 0.18. CRAN. <https://cran.r-project.org/package=extrafont> Grund, S., Robitzsch, A., & Lüdtke, O. (2021). mitml: Tools for Multiple Imputation in Multilevel Modeling. R package version 0.4-3. CRAN. <https://cran.r-project.org/package=mitml> Jones, K., & Subramanian, S. v. (2019). Developing multilevel models for analysing contextuality, heterogeneity and change using MLwiN 2.2 (3rd editio). University of Bristol.

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**The analyst’s conclusion:** The claimed relationship between the households’ total income and low versus high-caste village domination was identified. However, six outlying villages (6.67% of the whole sample) might be responsible for this observation because excluding these outliers results in no evidence supporting the claimed hypothesis.

#### **The analyst’s categorization of the result:**

The results show evidence for the relationship/effect as described in the claim provided in your task

#### **In Task 2, the analysts received the following instruction to conduct the analysis:**

You should not use District controls, Crop controls, Distance controls, Groundwater controls, Public goods controls Do not include Irrigation, credit, and tenancy variables, and their interaction term versions Do not use instruments for water buyer status.

#### **Here, the analyst reported the following steps and results of the analysis:**

I completed the task 1 the same way as task 2. Hence, I copy below the results of the previous analyses. The only addition is, according to instructions, specifying “a single, main result in terms of statistical families of z-, t-, F-, or <sup>2</sup> tests (or their alternative or non-parametric versions).” Hence, the main result is: Using the full dataset with outliers, type of villages’ caste domination is significantly related to households’ total income (  $\beta = 0.172$ , SE = 0.052, 95% CI[0.070, 0.273],  $t(1291) = 3.322$ ,  $p < 0.001$ ). Without six outlying villages, this difference ceases to be significant (  $\beta = 0.068$ , SE = 0.035, 95% CI[-0.001, 0.137],  $t(1291) = 1.946$ ,  $p = 0.052$ ).

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Intercept 28847435 5371 2110751 1453

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**Analysis language/software/code (optional to check):**

Task 1: R

Task 2: R

**Analysis codes:** <https://osf.io/mu4t7/>